

Bootstrap

Bootstrap Motivation

- Sometimes it is not possible to collect many samples from a population
- Sometimes it is not correct to assume a certain distribution for the population
- Goal: Assess sampling variation

Bootstrap Methods

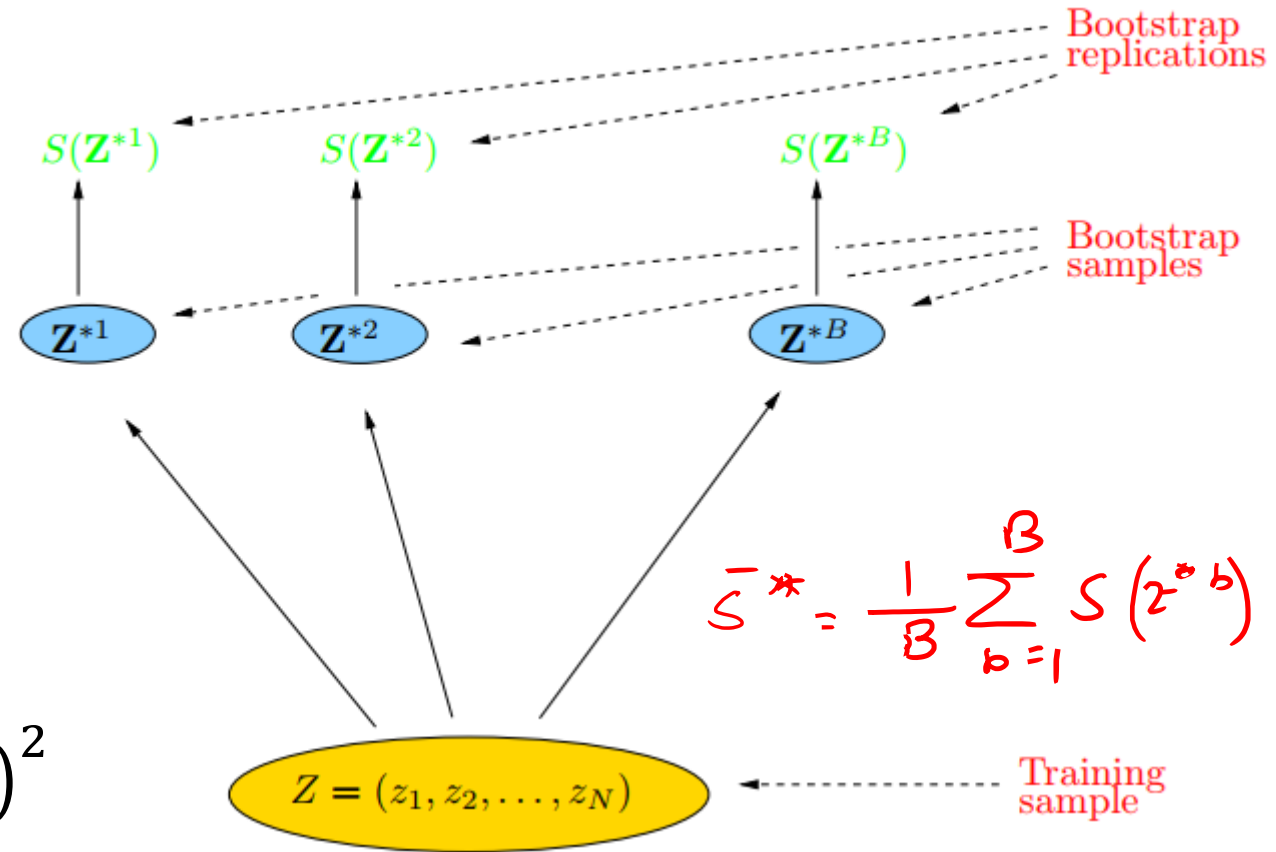
- The bootstrap is a general tool for assessing statistical accuracy.
- Lets denote the training set by $Z = (z_1, z_2, \dots, z_N)$ where $z_i = (x_i, y_i)$
- Randomly draw datasets with replacement from the training data, each sample the same size as the original training set.
- If you do it B times, producing B bootstrap datasets.
- Refit the model to each of the bootstrap datasets, and examine the behaviour of the fits over B replications.

Bootstrap Methods

- $S(Z)$ is prediction at some input Point of Z
- We can estimate any aspect of the distribution

$$\widehat{Var}[S(Z)] = \frac{1}{B-1} \sum_{b=1}^B (S(Z^{*b}) - \bar{S}^*)^2$$

- The prediction error: $\widehat{Err}_{boot} = \frac{1}{B} \frac{1}{N} \sum_{b=1}^B \sum_{i=1}^N \underline{L(y_i, \hat{f}^{*b}(x_i))}$
- $\hat{f}^{*b}(x_i)$ is the predicted value at x_i , from the model fitted to the b th bootstrap dataset.



Bootstrap Example

- Adapted from Fox (1997) “Applied Regression Analysis”
- Goal: Estimate mean difference between Male and Female
- Four pairs of observations are available:

Observ.	Male	Female	Differ. Y
1	24	18	6
2	14	17	-3
3	40	35	5
4	44	41	3

Mean = 2.75

Std Dev = 4.04

Sample	Y_1^*	Y_2^*	Y_3^*	Y_4^*	$\bar{Y}^*$
1	6	6	6	6	6.00
2	6	6	6	-3	3.75
3	6	6	6	5	5.75
..					
100	-3	5	6	3	2.75
101	-3	5	-3	6	1.25
...					
255	-3	3	3	5	3.5
256	3	3	3	3	3.00

Measuring Bias and Variance

- Create B bootstrap variants of the dataset
- Find out the predictions for each of the B bootstrap variants
- Compute the average of all the predictions $\overline{S}^* = \frac{1}{B} \sum_{b=1}^B S(\mathbf{Z}^{*b})$
- Bias: average across all models of the difference between the model prediction and the average prediction

$$Bias^2 = \left(f\left(z_{i_B}^{*b}\right) - \overline{S}^* \right)^2$$
$$\widehat{Var}[S(Z)] = \frac{1}{B-1} \sum_{b=1}^B \left(S(\mathbf{Z}^{*b}) - \overline{S}^* \right)^2$$